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Proterozoic palaeomagnetism of the mafic dyke swarms in the high-grade region of southern India

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Abstract

Palaeomagnetic investigations have been carried out on four major dyke swarms (38 sites) in the high-grade metamorphic region of the south Indian shield. A total of 37 sites (243 samples) in four dyke swarms have yielded Precambrian palaeomagnetic vectors after progressive AC and thermal demagnetisation: (1) the Agali–Anaikatti swarm bordering the Nilgiri Massif to the south; (2) the Tiruvannamalai swarm to the interior of the Pondicherry coast; (3) the Dharmapuri swarm towards the west of Tiruvannamalai in the interior of Tamil Nadu; and (4) a few dykes from a swarm in north Kerala on the west coast. Ages of the palaeomagnetic vectors are estimated from available K–Ar isotope dates on samples from a number of sites from all swarms and Rb–Sr isotope results from the older Agali–Anaikatti dykes. The vectors and their corresponding ages are as follows: $D=13.9°$, $I=27°$, $\kappa=94$, $\alpha_{95}=26$, pole = 75.7°N 153.2°E for the ~2000 Ma Agali–Anaikatti swarm ($N=2$ sites); $D=359.8°$, $I=9.2°$, $\kappa=51$, $\alpha_{95}=10$, pole = 79.7°N 101.1°W for the 1800–1750 Ma Dharmapuri swarm ($N=5$ sites); $D=217.2°$, $I=-50°$, $\kappa=103$, $\alpha_{95}=24$, pole = 51°N 130°E for the 1700–1650 Ma northern Kerala dykes ($N=2$ sites) and $D=106.9°$, $I=-77.6°$, $\kappa=42$, $\alpha_{95}=9.4$, pole = 18.8°S 125.2°W for the ~1650 Ma Tiruvannamalai swarm ($N=7$ sites). These results in conjunction with other available good quality data have been used to compile an apparent polar wander curve of India for the Proterozoic Eon. The palaeolatitudes derived from the data show that the Indian landmass was situated at mid-latitudes of the Northern Hemisphere during ~2000 Ma, at a low latitude near the Equator (~10°N) during ~1800–1750 Ma, close to the South Pole region (60°S) by ~1650 Ma, again near the Equator ~1000 Ma, near the North Pole (80°N) region at ~850–800 Ma, and moved to 10–30°N by 750 Ma.

1. Introduction

Palaeomagnetic data have been obtained from all the continents, including the Indian subcontinent, ever since plate tectonic concepts were established. Although the Palaeozoic Apparent Polar Wander Path (APWP) of Gondwana is generally well documented, the APWP of India in the Precambrian is not well established (Klootwijk, 1979). Judging from the available literature and the reviews of Indian palaeomagnetism (Klootwijk, 1979; Hargraves and Bhalla, 1983) it is clear that Precambrian data from southern India are plagued by the lack of direct isotopic age support and insufficient magnetic cleaning. Klootwijk (1979) commented on the suggestions for the Indian Precambrian APWP as "at best the optimistic oversimplifications". In view of this, we have undertaken a palaeomagnetic study of certain critical rock units in southern India. We have studied dyke rocks from the high-grade region of southern India primarily because our initial K–Ar isotope results (Radhakrishna et al., 1985) have shown that the dyke intrusions in this terrain were episodic since the Palaeoproterozoic.

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We have collected 251 samples from 38 sites in four major dyke swarms in the high-grade metamorphic region and have systematically determined stable palaeomagnetic vectors. We have also obtained K–Ar and/or Rb–Sr isotopic ages corresponding to each swarm vector. Only one site proved to be a Cretaceous dyke. Our results provide a substantial Precambrian palaeomagnetic record for southern India enabling the drawing of an apparent polar wander curve for India for the Proterozoic.

2. Geology of the dyke swarms

Numerous dyke intrusions occur within the high-grade crystalline basement of southern India (Fig. 1). Of these, the Agali–Anaikatti dykes (Fig. 2a) constitute an ENE–WSW swarm occurring along the Bhavani shear zone/lineament bordering the southern margin of the Nilgiri Massif (Fig. 1). The swarm is 20–25 km wide and extends from a few kilometres west of Agali to more than ~50 km towards the east in a general ENE–WSW direction. Some of the dykes show faint schistosity and subhorizontal lineations that correspond to shearing or a final phase of open folding without the dykes themselves being folded and can be traced in the strike direction. Still, there are dykes which are unmetamorphosed and devoid of these deformation features and these were chosen for the palaeomagnetic study. The Tiruvannamalai dyke swarm (Fig. 2b) occurs in the interior parts of Pondicherry on the east coast of India between longitudes 78°45′ and 79°45′E (see Fig. 1). The dykes in this area occur as two swarms in NE–SW and NW–SE orientations, with the latter being predominant. However, absence of cross-cutting relations precludes the establishment of a relative chronology between the dyke trends. The Dharmapuri dykes (Fig. 2c) occur between longitudes 77°45′ and 78°30′E, i.e., between the Agali–Anaikatti and the Tiruvannamalai swarms (Fig. 1). The Dharmapuri dykes strike NW–SE and constitute a relatively small swarm compared to the Agali–Anaikatti and the Tiruvannamalai swarms. The Agali–Anaikatti dykes occur within biotite and/or hornblende gneiss and migmatitic terrain; migmatite or charnockite is the country rock for the Tiruvannamalai dykes, whereas the dykes of Dharmapuri intruded the charnockite units. Dolerite dykes from the Dharmapuri and Tiruvannamalai areas are similar in their coarse-grained massive texture and black colour; the dolerites of Agali–Anaikatti are mostly medium-grained and greyish black in colour. Also the Tiruvannamalai and Dharmapuri dykes are much wider (generally >30–50 m and some times up to 100 m) than the Agali–Anaikatti dykes which are maximally only 5–6 m wide. The dykes in northern Kerala are also thin (5–6 m) and occur in NW–SE, NNW–SSE or NE–SW trends within the charnockite or tonalite gneiss country rock (detailed descriptions in Radhakrishna et al., 1991). The dykes in all areas have sharp chilled contacts and steep to vertical dips with the country rock.

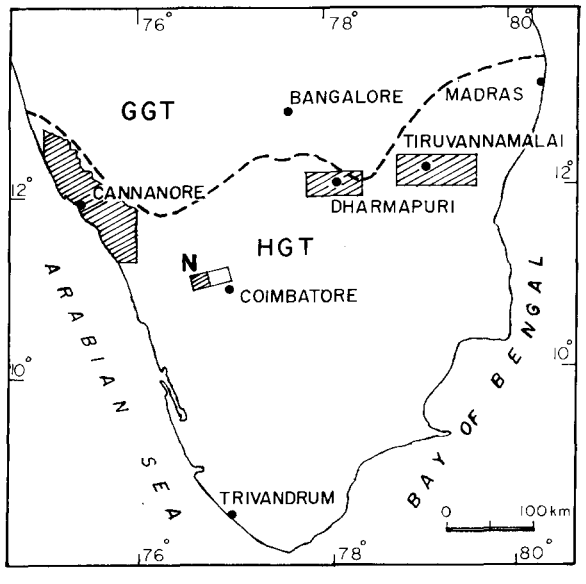


Fig. 1. Map of southern India showing the Precambrian dyke swarms of our study. *GGT* = granite–greenstone terrain; *HGT* = high-grade terrain; *N* = Nilgiri Massif.

Oriented block samples were collected (oriented with the magnetic compass) from nine sites (dykes) in the Agali–Anaikatti swarm (58 samples), ten sites (dykes) in the Dharmapuri swarm (66 samples) and fourteen sites (twelve dykes) in the Tiruvannamalai swarm (88 samples). Of the seventeen sites sampled in northern Kerala, only five sites (dykes) (39 samples) may be regarded as Precambrian in age.

3. Petrography and geochemistry

Detailed petrological and geochemical aspects of these dykes will be discussed elsewhere. Only the rel-

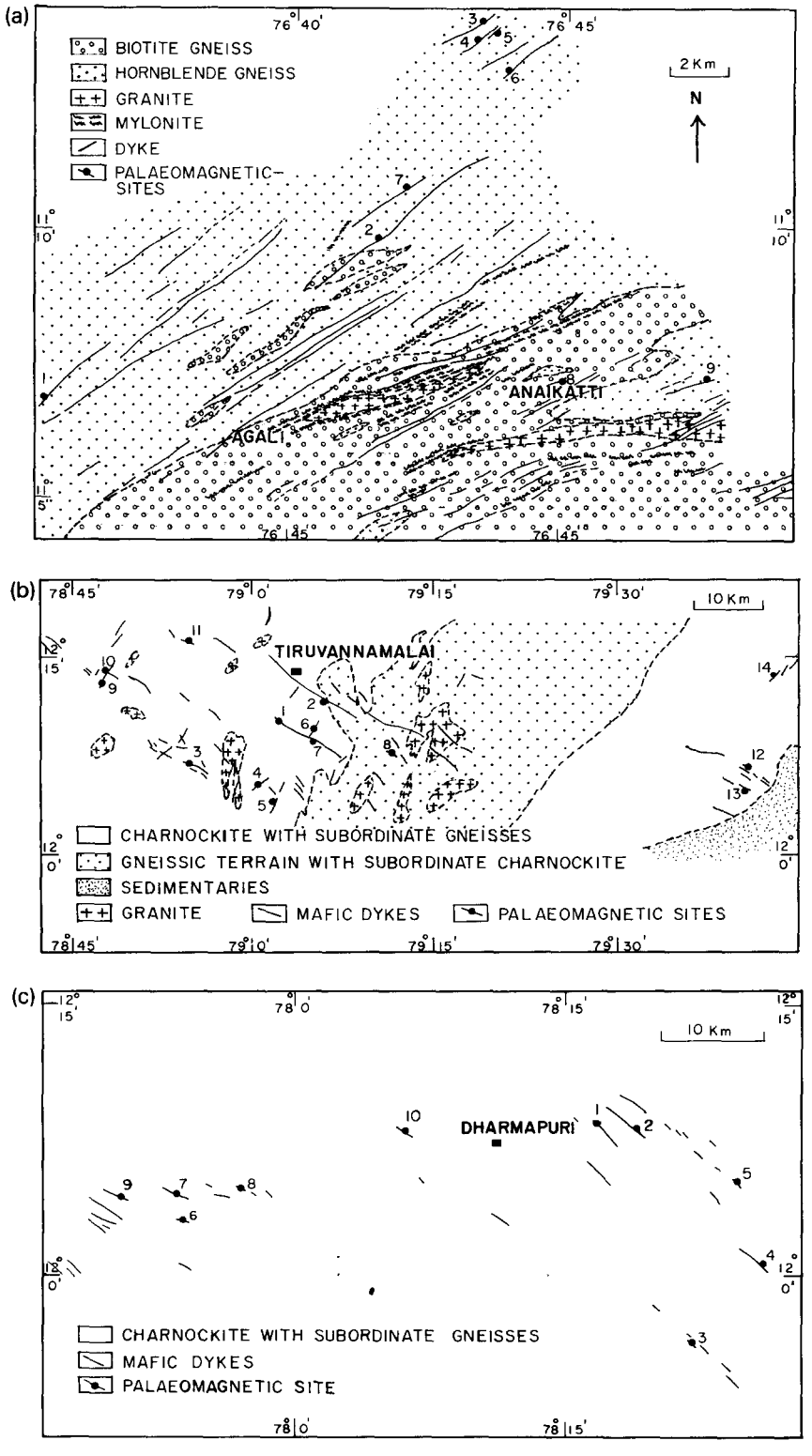


Fig. 2. Geological maps of (a) the Agali–Anaikatti dykes, (b) the Tiruvannamalai dykes (modified after Sugavanam et al., 1978), and (c) the Dharmapuri dykes showing the palaeomagnetic sites. Geological map of northern Kerala not shown since it was published earlier (Radhakrishna et al., 1991).

evant descriptions of the dykes of the palaeomagnetic sites are summarised here.

The unmetamorphosed Agali–Anaikatti dykes and the dykes of Dharmapuri, Tiruvannamalai and northern Kerala are generally similar in their petrographic characteristics. They contain clinopyroxene, plagioclase and Fe–Ti oxides as the essential mineral constituents. Olivine and minor orthopyroxene are the accessory subhedral–euhedral phenocryst phases. Their igneous origin is clearly evident from the diabasic textures and zoned plagioclase and clinopyroxene phenocryst phases. Clinopyroxene (augite) also occurs as microphenocrysts. Micropegmatitic intergrowths are present occasionally along the borders of large plagioclase crystals. Plagioclase is usually unaltered and seldom shows clouding. The mafic minerals (olivine and pyroxene) occasionally show minor alteration along grain boundaries or cleavage planes. Such altered olivines contain fine-grained magnetite garlanding the grain boundaries or as microveins across the grains. Pale green amphibole is the usual alteration product along grain boundaries and internal cleavages of pyroxenes. These alterations are judged to be deuteric. Among the oxides, magnetite with frequent exsolution lamellae of ilmenite constitute the Fe–Ti oxide dispersed throughout the rock as grains of subhedral and skeletal shape. Chrome spinel occurs in the more primary, less evolved rocks in association with olivine. Samples from the contact zones of some dykes are severely altered and the original textures are completely destroyed. Thus contact zones were not sampled for palaeomagnetic study. The geochemistry of these dykes indicates that they are olivine or quartz normative Fe-rich sub-alkalic tholeiites. There are no marked differences between the NW–SE and the NE–SW dolerites of Tiruvannamalai and both groups can be related to each other through fractionation. The elemental variations within each of the dyke swarms suggest that the dykes in each area form a consanguineous suite evolved through some degree of fractionation.

4. Palaeomagnetism and isotopic ages of the vectors

The oriented block samples collected in the field were reoriented in the laboratory and two/three cylindrical specimens of 2.54 cm in diameter and in length were obtained. The natural remanent magnetism (NRM) measurements were carried out using a Molspin-Minispin Spinner Magnetometer (MOLSPIN Ltd, UK). Progressive incremental AC and thermal demagnetisations were used to derive characteristic magnetic vectors using an AC shielded demagnetiser (MOLSPIN Ltd, UK) and a MMTD-1 thermal demagnetiser (Magnetic Measurements, UK). AC demagnetisations were performed on each sample in successive steps at close intervals (2.5 mT) until a stable end point was established. Stepwise thermal demagnetisations were performed on 61 representative samples incorporating every site. Thermal demagnetisations were performed in progressive steps at a 100°C interval until 450°C and then at 500, 525, 550, 580, 610, 640/650, 680 and 710°C. Orthogonal projections of magnetic remanence vectors (Zijderveld, 1967) and the standard stereographic projection methods were employed to delineate the characteristic vectors. The stable magnetic vectors thus obtained on each sample were used to compute the mean vectors of the site (dyke) and the dyke mean vectors were used to compute the mean characteristic vectors of each dyke swarm using Fisher's (1953) statistics.

A total of 39 samples from the four dyke swarms, including sites which yielded characteristic magnetic vectors, were dated by the K–Ar isotope method (Radhakrishna et al., 1995a). The Agali–Anaikatti dykes which are the oldest among the swarms were analysed for Rb–Sr and Sm–Nd isotopes (Radhakrishna et al., 1995b). The estimated ages of the characteristic vectors of each swarm are discussed on the basis of these results (see Radhakrishna et al., 1995a, for details).

4.1. Agali–Anaikatti dykes

Palaeomagnetism

Of the nine sites (dykes) sampled, one dyke site (site AG8) proved to carry a late Phanerozoic vector and yielded a similar K–Ar age; it is omitted from further discussion. The natural remanence at the NRM level of each sample shows large scatter between and within site vectors. Single magnetic characteristic vectors after AC demagnetisations are shown in Fig. 3. Orthogonal projections for AC and thermal demagnetisations of typical specimens are shown in Figs. 4 and 5.

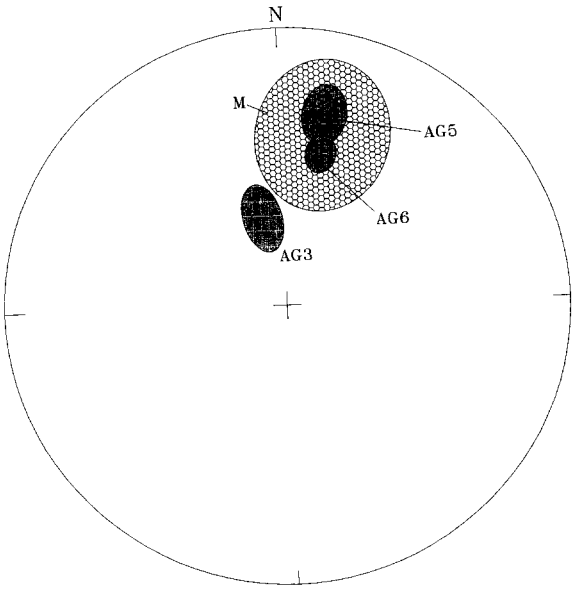


Fig. 3. Equal-area projections of characteristic magnetic vectors (as α_{95} ellipses) of the Agali–Anakatti dykes.

The dykes generally have NRM intensities of 1–5 A/m. On thermal demagnetisations, the directions of AG5 and AG6 site samples were intact until 525°C after initial removal of a soft component (Fig. 6). During AC demagnetisations, a small component was removed initially and thereafter the directions remained constant (for example, AG5A and AG6C; Fig. 4). Samples from these two sites showed a within- and between-site grouping after attaining stable vectors (Fig. 3). Samples from sites AG1, AG2, AG7 and AG9 either did not yield stable vectors or did not show within-site grouping. The sample vectors of these sites were unstable during thermal demagnetisations and, in a few samples, the vector directions (both D and I) approached closer to the directions of sites AG5 and AG6 (Fig. 6). In the case of the AG3 site, all except one sample (AG3B) stabilised after 10 mT during AC demagnetisations (Fig. 3), at $D=346.6°$ and $I=56.9°$ ($\alpha_{95}=14$; $\kappa=29$; $n=5$), a direction nearly identical to that of the 750-Ma Malani igneous suite ($D=354.5°$, $I=53.5°$; Klootwijk, 1979). However, during thermal demagnetisations, the directions showed a variable behaviour: sometimes they were stable up to 500°C or sometimes they continued to change with temperature and sometimes showed proximity to the vectors of the sites AG5 and AG6 between 250 and 500°C (Fig. 6). Further, a thermal event of this age is evident in the form of Late Precambrian (980–665 Ma) magmatic activity in southern India (cf. Kröner, 1991). Thus we suggest that the stable vectors as compared to Malani samples in site AG3 could be a clear case of remagnetisation. Because the AG5 and AG6 dyke sites show good within-site as well as between-site grouping and the vectors do not match with any other known directions of later thermal components or with any direction

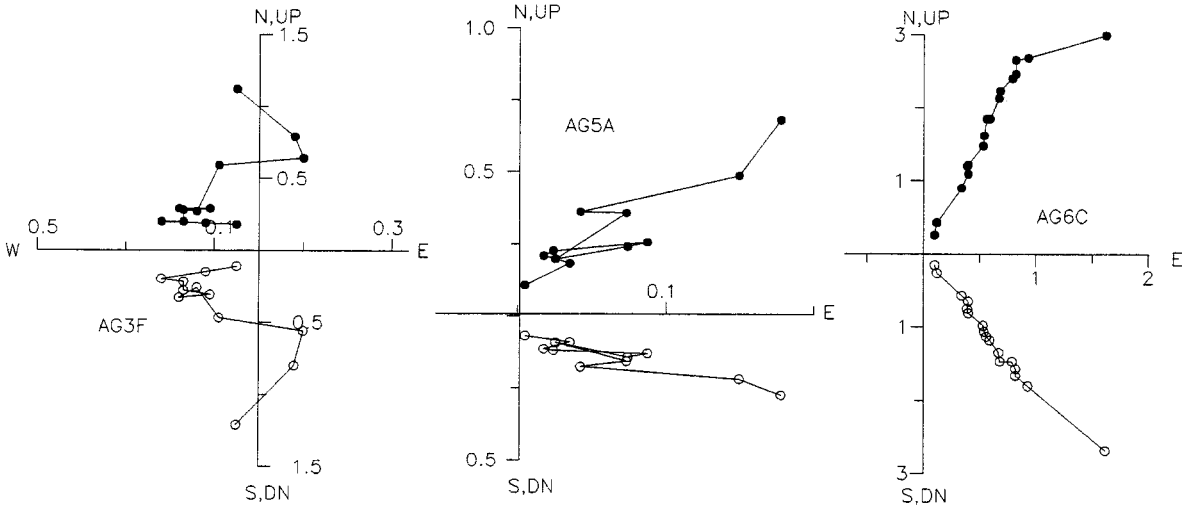


Fig. 4. Vector plots of AC demagnetisations on representative samples of the Agali–Anakatti dykes (in A/m). Open and closed symbols are vertical and horizontal projections, respectively.

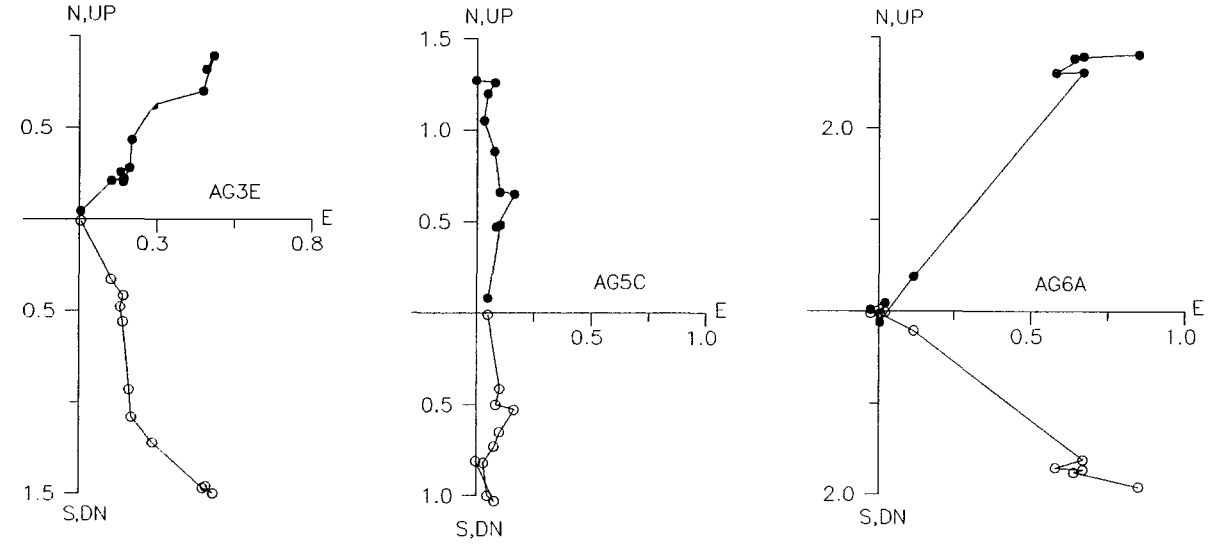


Fig. 5. Vector plots of thermal demagnetisations on representative samples of the Agali–Anakatti dykes (in A/m). Symbols are the same as in Fig. 4.

of the failed sites and because the site AG3 show continuous change with increase of demagnetising temperature, sometimes approaching closer to the AG5 and AG6 site vectors during thermal treatment, the vectors of the sites AG5 and AG6 are regarded as the primary remanent magnetic vectors of this Proterozoic dyke swarm. The palaeomagnetic results of this swarm are summarised in Table 1.

Age of the vectors

For five whole-rock samples (two are from the dykes corresponding to the AG2 and AG5 palaeomagnetic sites) and two light mineral (plagioclase) fractions duplicate Proterozoic K–Ar isotope ages were obtained (Radhakrishna et al., 1995a). Ages on three fresh samples cluster at ~1900–2000 Ma (1940±40 Ma, 1990±20 Ma, 2000±35 Ma; inverse variance weighted mean of 2000±20 Ma) while the other two whole rocks obtained ages of ~2500 Ma and ~4700 Ma. The plagioclase fractions yielded two lower but distinctive ages (~1100 Ma and ~1600 Ma). This large spread in the four samples is attributed to excess or loss of Ar. Rb–Sr, Sm–Nd whole-rock isotope data were available for six Proterozoic dyke samples of this area (Radhakrishna et al., 1995b). The Sm–Nd data show a good correlation (mean square weighted deviation, MSWD = 3.7) and the regression line gives an age of 3154±534 Ma (at 2σ error) which is older than the 2.5-Ga tectonothermal event known in the region. The correlation may reflect either mixing of old components or isolation of mantle sources and therefore the calculated age is geologically meaningless (Radhakrishna et al., 1995b). The Rb–Sr data generated a regression line (errorchron, Fig. 7) giving an age of 2369±398 Ma (at 2σ error) and this age is older than the actual emplacement age if minor older components are involved. Therefore, based on concordancy in the K–Ar ages of three different dykes and because the age is within the lower error limit of the Rb–Sr age, the characteristic palaeomagnetic vectors of the Proterozoic swarm of Agali are assigned an age of 2000±20 Ma.

4.2. Dharmapuri dykes

Palaeomagnetism

The mean vectors of natural remanence of each sample at the NRM level for the 10 sites of the Dharmapuri swarm show a large scatter within and between the sites. Orthogonal projections for the responses of typical specimens to AC and thermal demagnetisation are shown as Zijderveld plots (Figs. 8 and 9). The NRM intensities of most samples range from <1 to 6 A/m except a few samples of the D6 site (up to 19 A/m)

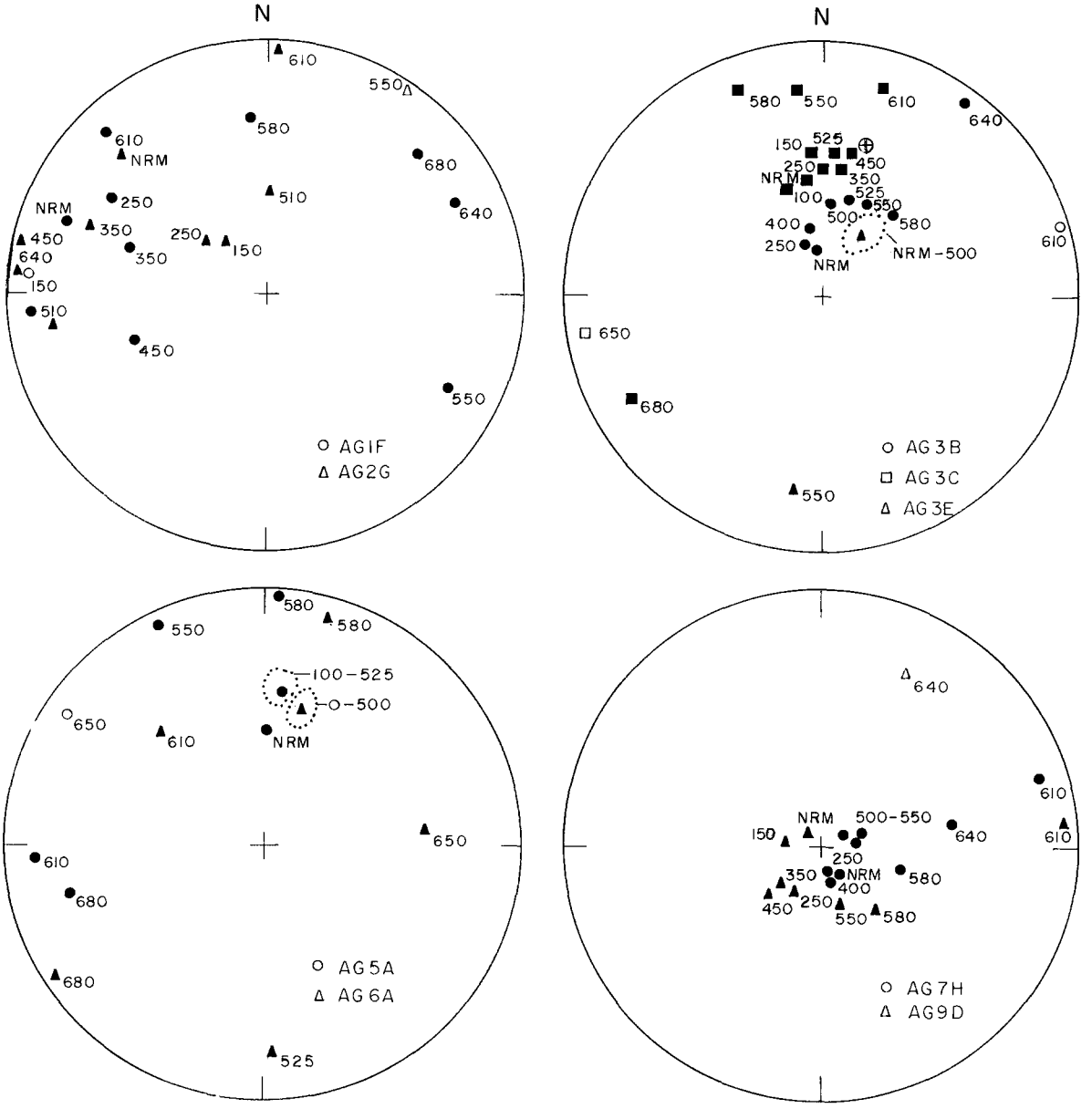


Fig. 6. Stereographic plot of stepwise thermal demagnetisation vectors of representative samples of the Agali–Anakatti dykes. Encircled plus-sign represents mean characteristic vector of the Agali–Anakatti swarm.

and a sample from the D9 site (D9E, 96 A/m). In nearly all samples the intensities fall to less than 10% of the original value in the first few steps of AC demagnetisation (<25 mT). Single characteristic magnetic vector components could be derived successfully in most of the samples after removal of soft (viscous) components of magnetisation in fields of 7.5–25 mT. Good within-site grouping is visible for six sites (D1, D4, D5, D7, D8 and D10; Fig. 10). The sites D1, D4, D5, D8 and D10 show good grouping between the sites with shallow inclinations; site D1 has a reverse polarity and the rest is normally magnetised. Sample vectors

from these sites do not show significant deviation on thermal demagnetisation and the intensities fall sharply or gradually to ~ 10% of the original intensity at 500–550°C. Stable vectors of four sites (D2, D3, D6, D9) are dispersed widely even for within-site distribution. Characteristic vectors of some samples from these sites yielded shallow vectors similar to the grouped site vectors, while in others they scatter towards steep positive inclinations. Some of the sample vectors are unstable during thermal demagnetisations suggesting that the scatter can be attributed to streaking due to later thermal events. Site D7 shows a within-site but distinct grouping with steep negative inclinations (Fig. 10). During thermal demagnetisations some samples (e.g. D7D) attain a direction close to the grouped site vectors between 550°C and 580°C, still retaining less than 20% of the remanence. Thus the magnetisation of the D7 site (dyke) may have been completely reset at a later thermal event probably related to the Tiruvannamalai dyke intrusions (described below). The vectors of this site are omitted in the calculation of mean swarm vectors.

The mean characteristic vectors of each site (dyke) and the mean swarm vectors are compiled in Table 2. Except one site, all the dykes yielded normal polarity with a very low 95% confidence interval. This probably indicates that the peak magmatic episode linked with the dyke intrusions occurred within a short time during a single polarity interval.

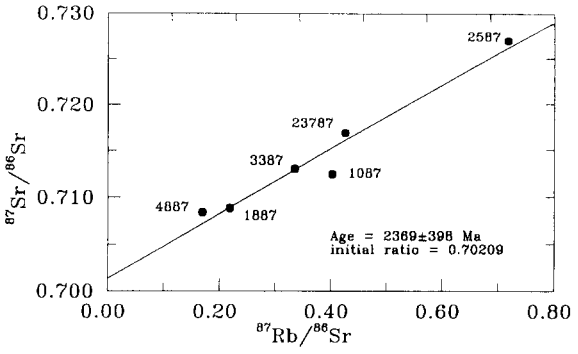


Fig. 7. $^{87}Rb/^{86}Sr$ vs $^{87}Sr/^{86}Sr$ plot of the Agali–Anakatti dykes. Sample 1087 not included in age calculation.

Age of the vectors

K–Ar isotope measurements have been made on one sample from each of the sites D1, D2, D3, D4, D5, D7, D9, D10 and on three samples from site D8 (Table 2). Except for two ages from the ungrouped sites D3 (1300 Ma) and D9 (2600 Ma), which are attributed to excess/loss of Ar, the dates of the other samples fall into three age brackets: 1800–1700 Ma, 1600–1500 Ma and 1500–1400 Ma. Of the three samples from the D8 site, two samples fall in the 1600–1500 Ma cluster, while one falls into the 1800–1700 Ma cluster. The problem is to decipher whether the Dharmapuri dykes were intruded at 1800–1700 Ma or 1500 Ma. However, dyke D7, which is palaeomagnetically similar to the

Table 1
Palaeomagnetic results of the Agali–Anaikatti dykes

Site No.	Trend	Lat. (°N)	Long. (°E)	N	N_s	D	I	κ	α_{95}	λ	ϕ
@AG1	NNE–SSW	11.12	76.51	8	1	–	–	–	–	–	–
@AG2	NNE–SSW	11.16	76.78	7	0	—no stable vector—					
+AG3	NNE–SSW	11.23	76.81	6	5	346.6	56.70	9	14.4	62.6	36.8
@AG4	NNE–SSW	11.22	76.81	5	3	—no grouping—					
*AG5	NNE–SSW	11.22	76.82	5	4	13.25	21.10	122	8.3	77.0	166.8
*AG6	NNE–SSW	11.21	76.83	6	6	14.59	32.89	130	5.9	74.4	139.6
@AG7	NNE–SSW	11.18	76.78	7	0	—no stable vector—					
@AG9	NNE–SSW	11.12	77.04	6	5	—no grouping—					
Mean swarm vector					2		13.9	27.0	94	26.1	

Palaeomagnetic pole: 75.7°N 153.2°E; palaeolatitude: 24.6°N

N = number of samples studied from the site; N_s = number of samples that yielded stable vectors; D and I are the declination and inclination (in degrees), respectively, of the characteristic remanent magnetic vectors; κ = precision parameter; α_{95} = radius of the 95% confidence cone about the mean direction; λ and ϕ are respectively the latitude and longitude of VGP; * = site mean values included to calculate mean swarm vectors, and @ = site not considered for mean calculation either because it failed to yield stable end point or grouping; + = an anomalous or a distinct direction that is not considered for mean swarm vectors.

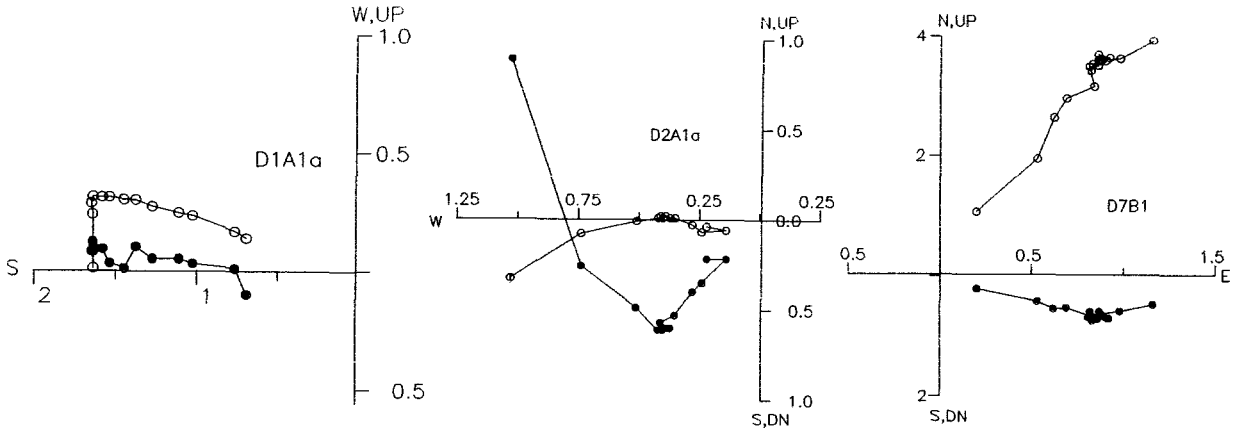


Fig. 8. Vector plots of AC demagnetisations on representative samples of the Dharmapuri dykes (in A/m). Symbols are the same as in Fig. 4.

1650-Ma Tiruvannamalai dykes (below), falls into the 1500–1400 Ma time span, probably because of loss of some Ar. That is, either the D7 dyke is an isolated dyke in the Dharmapuri area coeval with the Tiruvannamalai dykes or it represents a remagnetised direction in the Dharmapuri area at the time of the Tiruvannamalai dyke intrusion (i.e., ~1650 Ma). However, we suggested earlier that the D7 site is likely to have remagnetised at ~1650 Ma and the Dharmapuri vectors, which, if younger, are likely to be traced in the nearby Tiruvannamalai area, are totally unnoticed in the Tiruvannamalai swarm. Also the Dharmapuri dykes are positioned between the ~2000-Ma Agali–Anaikatti dykes and the ~1650-Ma Tiruvannamalai dykes on the polar wander curve and palaeolatitude figures (next section). Thus an inverse variance weighted mean age of 1760±30 Ma, calculated from the three ages between 1800 and 1700 Ma (Radhakrishna et al., 1995a), is more tenable for these dyke vectors.

4.3. Tiruvannamalai dykes

Palaeomagnetism

The natural remanences at the NRM level for all fourteen sites show a large scatter within and between the site samples as in the case of other Proterozoic swarms. A few examples of AC and thermal demagnetisation responses on samples are shown in Figs. 11 and 12. The stable magnetic vectors are plotted in Fig. 13. Intensities are generally <10 A/m but values of

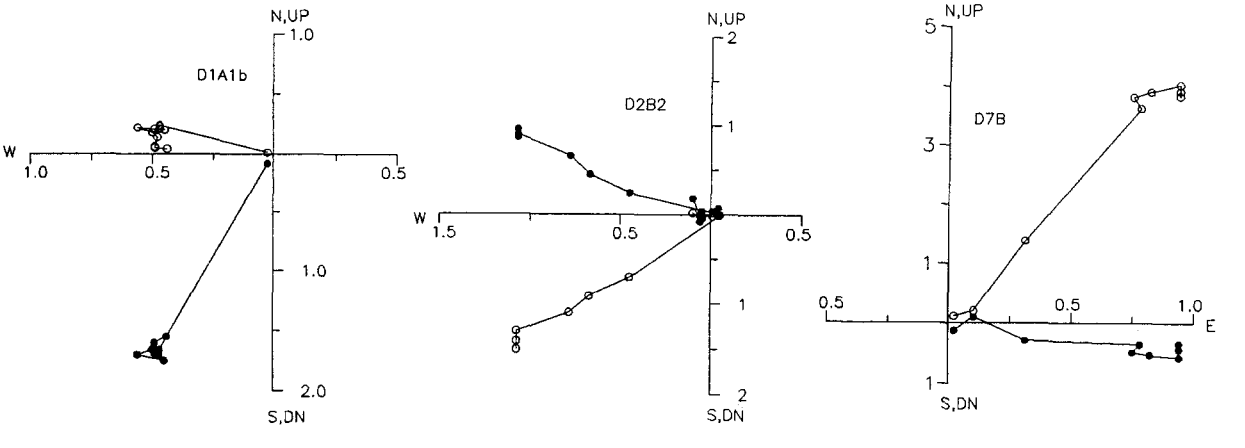


Fig. 9. Vector plots of thermal demagnetisations on representative samples of the Dharmapuri dykes (in A/m). Symbols are the same as in Fig. 4.

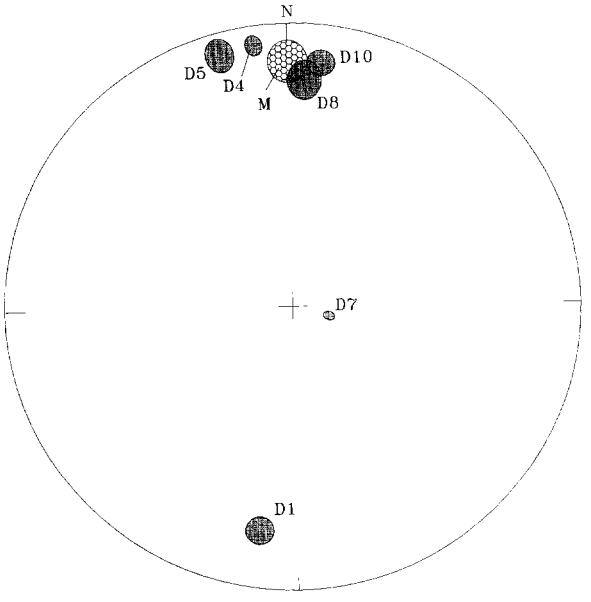


Fig. 10. Equal-area projections of characteristic magnetic vectors (as α_{95} ellipses) of the Dharmapuri dykes.

~30–40 A/m are not uncommon. Sites T1 and T6 recorded intensities of >100 A/m. On AC demagnetisations many of the samples show gradual changes in the NRM and single characteristic magnetic vectors could be derived for the majority of the samples (Fig. 13); still a few samples did not attain stable vectors even after erasing >90% of the original intensity (up to 60 mT). During thermal demagnetisations, the samples generally retained the initial directions, which otherwise were erased in fields of <20 mT by AC demagnetisations, at least up to 500°C. Titanomagnetite is the main magnetic carrier in the samples and thermal demagnetisations are not successful in obtaining primary vectors.

Among the site (dyke)-characteristic vector data, sites T3, T5, T7, T8, T9, T12 and T13 showed good grouping after AC demagnetisations below the 20 mT increment. Sites T3, T7 and T8 show good within-site as well as between-site grouping with steep negative inclinations and near-easterly declinations (Fig. 13). T9 shows strong within-site grouping in the eastern hemisphere with shallow inclinations (Fig. 13). Incidentally, this direction closely matches with the reverse vector of Group I dykes (Hargraves and Bhalla, 1983), which is clearly post-Cuddapah, and the direction was assigned an age of 980±110 Ma. Sites T12 and T13 have also well grouped stable directions upon AC demagnetisation, but their directions are distributed between the mean site directions of site T9 and sites T3 (or T7 and T8). Many characteristic vectors of the samples from T2, T4, T6, T10 and T11 are comparable to those of sites T3, T7 and T8. However, a few site samples do not congregate into a cluster and spread away in isolation between the site T3 (or T7 and T8) and site T9 clusters, clearly reflecting streaking from

Table 2
Palaeomagnetic results of the Dharmapuri dykes

Site No.	Trend	Lat. (°N)	Long. (°E)	N	N_s	D	I	κ	α_{95}	λ	ϕ	K–Ar age (Ma)
*D1	NW–SE	12.14	78.27	7	7	189.62	−13.95	70	7.3	68.5	231.3	1780±25
@D2	NW–SE	12.13	78.32	8	8	—no grouping—						1605±20
@D3	NW–SE	11.94	78.36	7	5	—no grouping—						1350±10
*D4	NW–SE	12.01	78.42	7	7	352.70	4.50	150	4.9	77.9	295.6	1495±25
*D5	NW–SE	12.09	78.39	7	4	345.05	5.40	119	8.4	72.5	317.3	1565±20
@D6	NW–SE	12.04	77.83	7	4	—no grouping—						
+D7	NW–SE	12.08	77.89	6	6	105.76	−75.47	168	5.2	−17.9	230.2	1460±25
*D8	NW–SE	12.08	77.94	6	6	4.17	13.54	57	10.2	83.4	219.3	1805±25
@D9	NW–SE	12.08	77.84	6	6	—no grouping—						2610±20
*D10	NW–SE	12.33	78.10	5	5	7.84	8.15	122	6.9	78.7	214.0	1710±20
Mean swarm vector					5		359.8	9.2	51	10.8		

Palaeomagnetic pole: 79.7°N 101.1°W; palaeolatitude: 10.7°N

All notations as in Table 1.

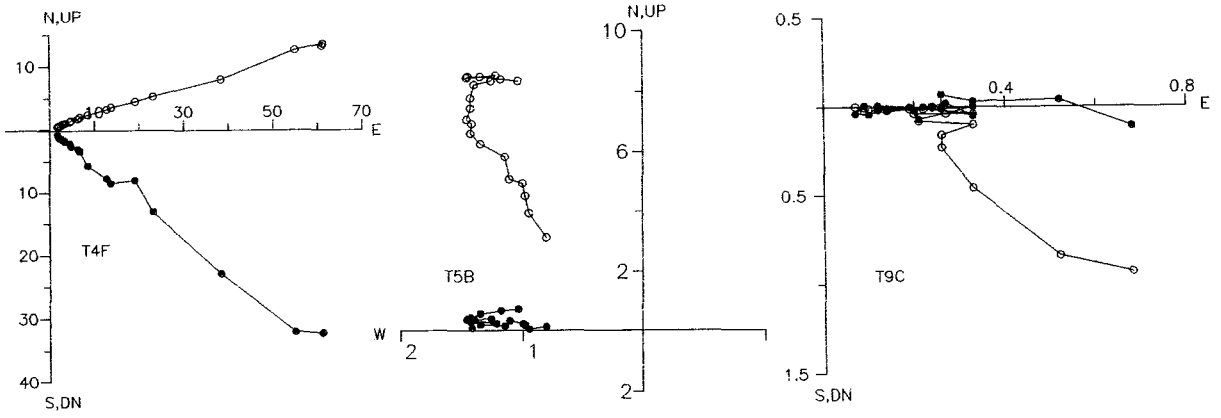


Fig. 11. Vector plots of AC demagnetisations on representative samples of the Tiruvannamalai dykes (in A/m). Symbols are the same as in Fig. 4.

one group to the other. Here the most complex problem is to delineate the primary and secondary vectors. One possibility is that both clusters, observed at shallow and steep inclinations, may belong to two distinct magmatic events and thus represent two independent primary vectors. However, the dykes of these distinct vectors have similar geochemical characteristics that can be linked genetically (Radhakrishna and Joseph, 1995) and the K–Ar ages of these dykes cluster at ~ 1650 Ma relating them precisely to a single igneous event. When these points are coupled with a direction in the Tiruvannamalai dykes that can be attributed to a late thermal event, ~980 ± 110 Ma Group I dykes (Hargraves and Bhalla, 1983), the magnetic vectors with shallow inclination can be argued as being due to the complete resetting of magnetisation. Thus the vectors with steep inclination in the sites T3, T7 and T8 can be regarded as primary vectors. The sample directions of site T12 or any similar direction from these dykes falling between the two groups may be attributed to streaking. Following this argument for sites T2, T4, T10 and T11, only the sample vectors of steep inclinations which show grouping with the T3, T7 and T8 stable vectors have been treated as primary. Site T5 shows good within-site grouping with steep negative inclinations and a northwesterly declination ($D = 307°, I = -76.7°$, $\alpha_{95} = 4.1$). In view of its indistinguishable age and geochemical characteristics when compared with the

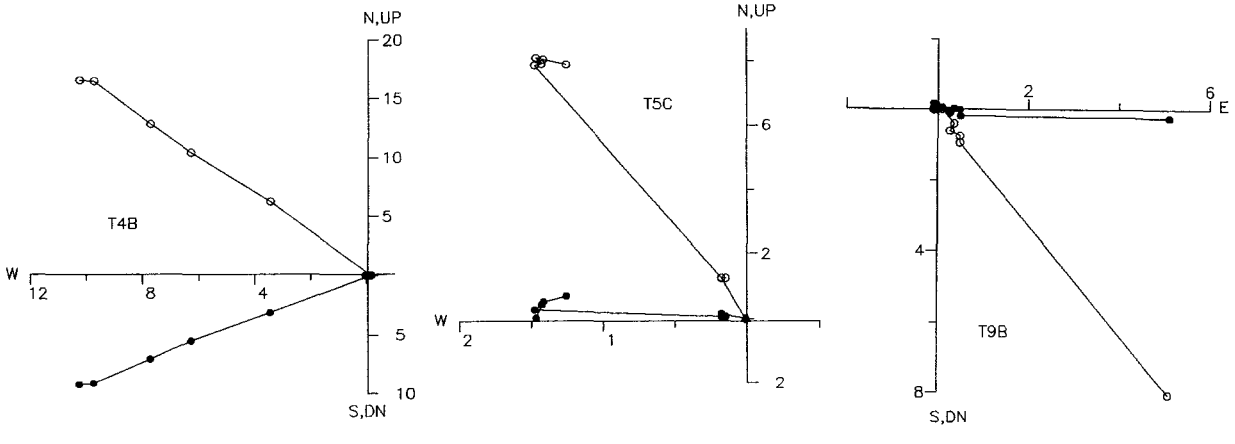


Fig. 12. Vector plots of thermal demagnetisations on representative samples of the Tiruvannamalai dykes (in A/m). Symbols are the same as in Fig. 4.

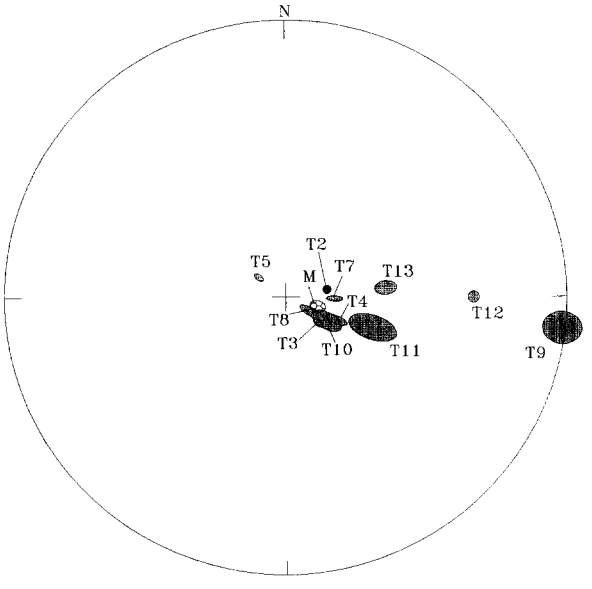


Fig. 13. Equal-area projections of characteristic magnetic vectors (as α_{95} ellipses) of the Tiruvannamalai dykes.

other dykes in the swarm, its large longitudinal shift to the northwest is considered as due to secular variation. Therefore, seven sites in total (T2, T3, T4, T5, T7, T8 and T10) have yielded primary magnetic vectors ($D = 106.9°$, $I = -77.6°$). The mean dyke (site) vectors and the mean swarm vectors are compiled in Table 3.

Earlier, Venkatesh et al. (1986) reported three different vectors from the Tiruvannamalai dyke swarm corresponding to our mean vectors of sites T3 (T7 and T8), T5 and T12 (or T13) and related them to distinct intrusive phases. However, in the light of the above discussions we suggest that only the direction of group C ($D = 133°$, $I = -75°$) can be regarded as primary and others can be attributed to streaking or complete resetting of the magnetism.

Age of the vectors

K–Ar isotope dates were obtained on nine sites (dykes) of the fourteen palaeomagnetic sites (Table 3). Out of nine ages, six are highly concordant and clearly define an event at ~1650 (inverse variance weighted mean of 1650 ± 10 Ma). Three ages, which are older, spread between 2200 and 2500 Ma. Possible interpretations for these data are that the dykes could have intruded at ~1650 Ma and the older ages are due to excess Ar in the sample. An alternative explanation could be a pre-2500 Ma emplacement with a regional event (? metamorphism) at 1650 Ma. Since the dykes are fresh and devoid of metamorphism, an age of 1650

Table 3
Palaeomagnetic results of the Tiruvannamalai dykes

Site No.	Trend	Lat. (°N)	Long. (°E)	N	N_s	D	I	κ	α_{95}	λ	ϕ	K–Ar age (Ma)
@T1	NW–SE	12.15	79.03	5	3	—no grouping—						
*T2	NW–SE	12.20	79.08	6	2(1)	79.7	−74.7	–	–	−5.8	230.7	1630 ± 30
*T3	NW–SE	12.09	78.92	7	7	129.7	−73.5	481	2.8	−29.9	232.0	
*T4	NE–SW	12.06	79.01	7	7(4)	114.4	−67.9	92	9.6	−24.6	219.8	1650 ± 30
*T5	NE–SW	12.05	79.03	7	7(6)	306.5	−76.7	275	4.1	3.4	279.2	1650 ± 30
@T6	NE–SW	12.05	79.08	5	3	—no grouping—						1660 ± 30
*T7	NW–SE	12.05	79.08	7	7	92.1	−70.7	69	7.3	−11.0	223.3	2550 ± 30
*T8	NW–SE	12.11	79.10	6	3	121.0	−78.1	109	11.9	−29.9	232.0	
+T9	NE–SW	12.22	78.80	7	6	96.6	−0.8	111	6.4	−6.5	167.8	2210 ± 20
*T10	NW–SE	12.22	78.80	6	5(2)	120.7	−72.1	57	33.7	−26.7	227.3	2270 ± 40
+T11	NW–SE	12.27	78.91	7	5(3)	109.6	−55.8	45	18.5	−22.9	203.4	
+T12	NW–SE	12.08	79.66	7	7	90.1	−23.1	550	2.8	−2.6	181.4	1670 ± 20
+T13	NW–SE	12.06	79.66	6	6	84.3	−51.1	74	7.9	−1.6	201.9	
@T14	NE–SW	12.22	79.72	5	4	—no grouping—						1640 ± 20
Mean swarm vector					7		106.9	−77.6	42	9.4		

Palaeomagnetic pole: 18.8°S 125.2°W; palaeolatitude: 59.2°S

All notations as in Table 1. Number of samples considered for primary vector mean calculations are given in brackets.

Table 4
Palaeomagnetic results of the dykes of northern Kerala

Site No.	Trend	Lat. (°N)	Long. (°E)	N	N_s	D	I	κ	α_{95}	λ	ϕ
@C2	NE–SW	12.15	75.43	7	0	—no stable vector—					
@C3	NW–SE	12.15	75.45	7	0	—no stable vector—					
@C4	NE–SW	12.13	75.45	7	0	—no stable vector—					
*C6	NNW–SSE	12.36	75.30	7	3	208.40	−50.85	14	34.9	57.6	124.4
*C18	NNW–SSE	12.30	75.20	11	10	225.50	−48.50	48	7.2	44.0	135.0
Mean swarm vector					2		217.16	−49.99	103	24.8	
Palaeomagnetic pole: 50.9°N 130.5°E; palaeolatitude: 40°S											

All notations as in Table 1.

Ma could be assigned to the actual intrusion of this swarm.

4.4. Northern Kerala

Among the thirteen sites (dykes) of our recent palaeomagnetic study, a Proterozoic age may be attributed only to four sites (C2, C3, C4 and C6). None of the samples from C2, C3 and C4 yielded stable vectors, and a sample from dyke C3 has yielded a Proterozoic (2100 Ma) K–Ar isotope age. Three out of seven samples from site C6 showed stable vectors at fields of < 15 mT. On thermal demagnetisations, the directions remain intact up to over 500°C and titanomagnetite is the main magnetic carrier. The stable vectors obtained on three samples upon AC demagnetisations attained a mean direction of $D = 208.4°$ and $I = -50.8°$ with low precision ($\kappa = 14$; $\alpha_{95} = 34$). This mean is close to the mean direction of the Palachal dyke vectors ($D = 225.5°$, $I = -48.5°$) reported earlier (Radhakrishna et al., 1986). The directions are listed in Table 4 and are shown in Fig. 14. Earlier duplicate K–Ar ages of 1660 ± 25 and 1668 ± 25 Ma have been reported for the Palachal dyke (Radhakrishna et al., 1986, 1991). Therefore the direction can be assigned an age between 1700 and 1650 Ma. Such an age is consistent with its intermediate position between the 1760 ± 30 Ma Dharmapuri dykes and the 1650 ± 10 Ma Tiruvannamalai dykes on the polar wander curves and palaeolatitude diagram discussed below.

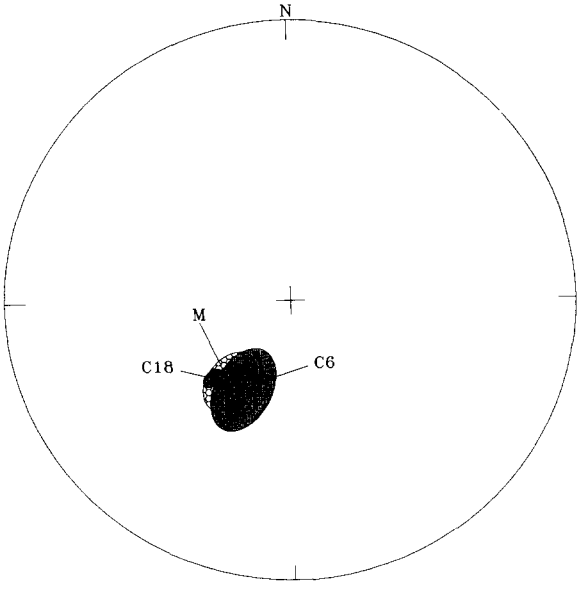


Fig. 14. Equal-area projections of the characteristic palaeomagnetic vectors (as α_{95} ellipses) of the Proterozoic dykes of northern Kerala. C18 represents the Palachal dyke from Radhakrishna et al. (1986).

5. Discussion and the Proterozoic APWP of India

Klootwijk (1979), while reviewing the Indian Precambrian palaeomagnetism, stated "A substantial amount of Precambrian palaeomagnetic data for the Indo–Pakistan subcontinent has been accumulated, but only a very few of the formations studied have been radiometrically dated. For most of the results only rough relative ages can be assigned, based mainly on superposition and lithological correlation... Because of the age uncertainty of most of the poles, the proper location of the Precambrian poles along an apparent polar wander path becomes rather hazardous exercise.

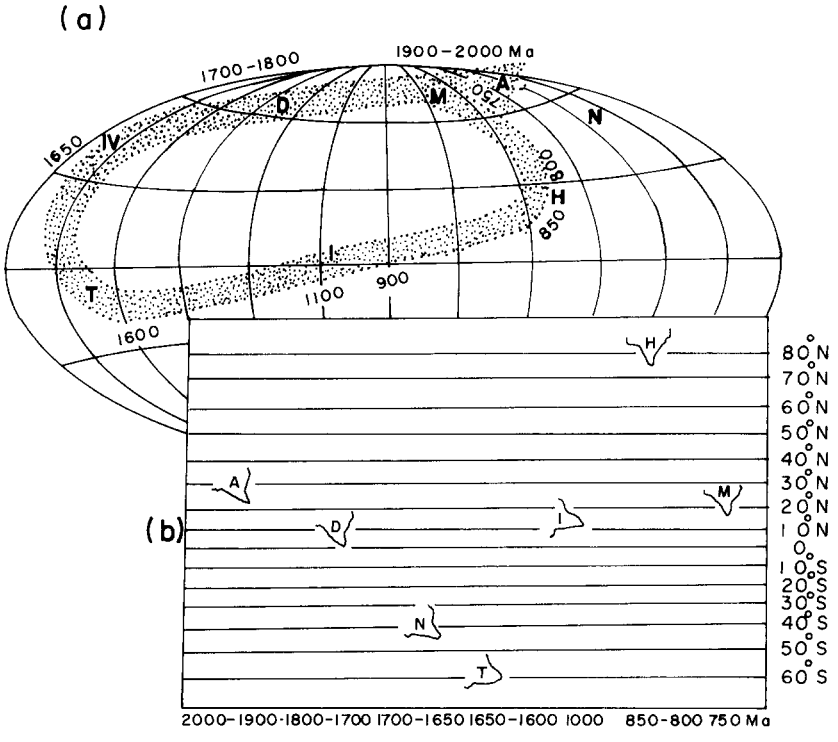


Fig. 15. (a) Proterozoic polar wander curve of India, and (b) its latitudinal movements. The poles and the palaeolatitudes in (a) and (b) are denoted as follows: A = Agali; D = Dharmapuri; T = Tiruvannamalai; N = northern Kerala; I = Group I dykes; IV = Group IV dykes (Hargraves and Bhalla,1983); H = Harohalli alkaline dykes; M = Malani rhyolite. For details see text.

Of course, one can place the various pole positions in sequence according to the relative ages assigned to their respective formations, but with the present scanty data this is equally hazardous." Hargraves and Bhalla (1983) stated that it is premature to attempt to draw a temporal APWP path through the data because none of the vectors have supporting age data. In addition, palaeomagnetic data from the Indian subcontinent are stated to be of variable quality; especially none of the results from southern India constitute the class A category of Klootwijk (1979), which incorporates data which (i) have supposedly been cleaned of any secondary component, (ii) are based on a sufficient number of samples to average the secular variation effects, and (iii) are of reasonable accuracy ($\alpha_{95} \leq 20$). In the light of these criteria, our present results are a significant contribution towards an attempt to draw the Proterozoic apparent polar wander curve because these results include different vectors of diverse age which have been isotopically dated.

Similar significant results were obtained recently by Dawson and Hargraves (1994) on the Harohalli dykes which were dated at 2370 ± 230 Ma and ~850–800 Ma (Ikramuddin and Stueber, 1976). The ~850–800 Ma vector can be considered primary. However, the results on the older dyke swarm appear to be uncertain. The mean palaeomagnetic vector on the older dykes ($D = 79.6°$, $I = -82.3°$) are closely comparable to the mean direction of the Tiruvannamalai dykes with a relatively better constrained age of ~1650–1600 Ma. If the older Harohalli dykes are over ~2000 Ma, one possibility is that in the light of our results on the Tiruvannamalai dykes, these dyke vectors were completely reset at ~1650 Ma when the Tiruvannamalai dykes intruded. Alternatively, the poorly constrained older age may have resulted due to whole-rock Rb/Sr correlation imparted by the involvement of some old component in the petrogenesis rather than reflecting the actual time of dyke intrusions, a case which needs further investigation. In either case, we reaffirm the view

that the age of this magnetisation is very uncertain (Dawson and Hargraves, 1994). Interestingly, the vectors of site 14 ($D = 246.9°$, $I = -35°$; Dawson and Hargraves, 1994) are closely similar to our reported Proterozoic vectors from northern Kerala. Thus our Proterozoic dyke vectors of northern Kerala together with the site 14 vectors of the Harohalli dykes may constitute a group representing a 1700–1650 Ma palaeomagnetic vector of southern India. Apart from these results, a palaeomagnetic record was reported for the Malani suite which was dated at 750 Ma (Klootwijk, 1979).

Based on the above data, we attempted to construct the apparent polar wander curve and the latitudinal position of India during the Proterozoic Eon (Fig. 15). The poles of the Agali–Anaikatti dykes are the oldest (2030 ± 20 Ma), fall near to the present-day pole, and India was situated to the north of the Equator at $\sim 25°$ latitude. The poles of the Dharmapuri dykes, to which we assigned an age of 1760 ± 30 Ma, are little away from the Agali–Anaikatti pole; the southern Indian craton moved to the south and was situated in the low latitudes of the Northern Hemisphere. By the time of the Tiruvannamalai dyke intrusion ($\sim 1650 \pm 10$ Ma) the pole shifted a little to the south towards the equatorial region and the craton reached the southern polar region ($\sim 60°$S latitude). The Proterozoic (1700–1650 Ma) dykes from northern Kerala occupy an intermediate position between the Dharmapuri and Tiruvannamalai poles. Following 1650 Ma, the Indian landmass migrated to the polar latitudes (86°N) of the Northern Hemisphere at the time of the alkaline dyke intrusions of Harohalli (850–800 Ma). The Group I dykes (Hargraves and Bhalla, 1983) were assigned an age of $\sim 980 \pm 110$ Ma, the pole lying between the locations of the Tiruvannamalai and Harohalli alkaline dyke poles, and India occupied a low-latitude position in the Northern Hemisphere. The Wajrakarur kimberlite poles, which were assigned an age of ~ 1150 Ma (cf. Hargraves and Bhalla, 1983), also plot in the vicinity of the 1100–1000 Ma strip of the curve. However, we suspect that this direction may represent some streaking because its vectors are similar to our T12 and T13 vectors which are suggested to be streaking directions. At the time of the Malani igneous activity (~ 750 Ma), the Indian pole shifted to a position similar to the present-day pole and the Indian landmass moved to 10–30°N. Thus the apparent polar wander curve designed here for the Proterozoic satisfies the existing palaeomagnetic record of India. However, more precise age data on the vectors falling on or close to the curve and additional information on any of the rock suites of intermediate ages, particularly between 1650 and 850 Ma, would improve considerably the proposed apparent polar wander curve.

5. Conclusions

The present status of the palaeomagnetic record of southern India is not of adequate quality for constructing a detailed APW curve of India, especially for the Precambrian Eon. Here, we have compiled mean palaeomagnetic vectors for all major dyke swarms in the high-grade region of southern India together with isotope ages for the dykes. This constitutes the best data base currently available.

The Precambrian vectors derived from our study correspond to episodic Proterozoic igneous intrusions at ~ 2000 Ma, 1760 Ma, 1700–1650 Ma and 1650 Ma. Based on these results and the palaeomagnetic record of the Harohalli alkaline dykes (~ 850–800 Ma; Dawson and Hargraves, 1994) and the Malani igneous suite (~ 750 Ma; cf. Klootwijk, 1979) an apparent polar wander curve of India is proposed for the Proterozoic. The palaeolatitudes show that India was situated in the mid-latitudes ($\sim 40°$N) of the Northern Hemisphere during 2000–1900 Ma, in the high southern polar latitudes (60°S) at ~ 1650 Ma, then migrated to the northern polar region ($\sim 80°$N) by 850–800 Ma, and reached low latitudes (10–30°N) by 750 Ma. We agree that the curve is still poorly constrained and can be improved as more high quality data are forthcoming. Further work towards filling the gaps and more precise isotope ages for the presently reported vectors are recommended for improving the proposed polar wander curve of Proterozoic India.

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